GOODWE
SOLAR ACADEMY

2023 TECHNICAL DOCUMENT

GoodWe smart charging Solution

Introduction

With the popularization of electric vehicles, the demand of combine the residential PV or storage system to control the charging of electric vehicles is more and more intense. GoodWe provides the smart charging solution for all the grid-tied and storage inverters.

EV Charger

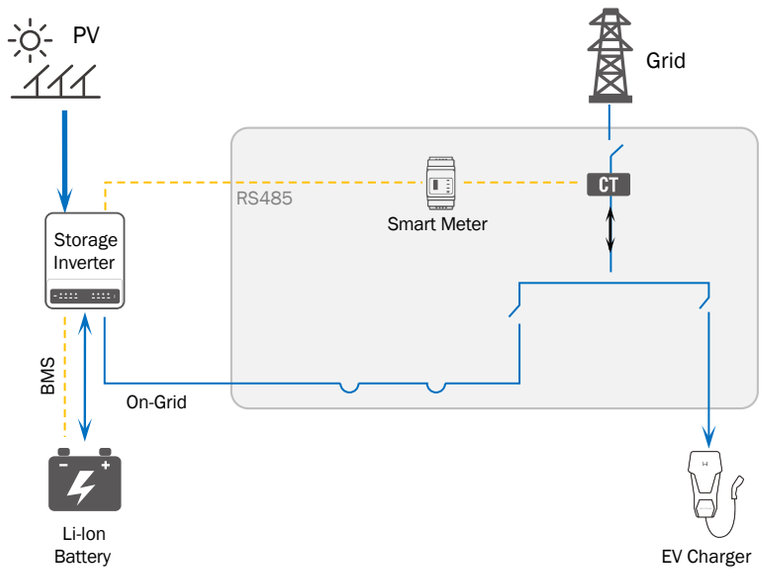
GoodWe HCA series EV charger includes 7kw for single phase and 11kw/22kw for three phase.



	GW7K-HCA	GW11K-HCA	GW22K-HCA
Ouptut power(W)	7000	11000	22000
Output voltage(Vac)	230	400	400
Output current(A)	32	16	32

GoodWe EV charger integrated multiple protections, such as overvoltage and undervoltage protection, overload protection, ground fault protection, lightning protection. Also, GoodWe EV charger have the IP65 protection degree to make sure it can work both indoors and outdoors. Regard to the communication, GoodWe EV charger have 2 communication dongles, wifi and Bluetooth.

Smart charging System connection



GoodWe residential EV charger system includes pv panels, grid-tied inverter or hybrid storage inverter, Li-Ion battery, smart meter with CT, other home loads , EV charger and the grid.

EV Charger Mode Introduction

GoodWe EV Charger has 3 different modes.

1. Fast charge mode

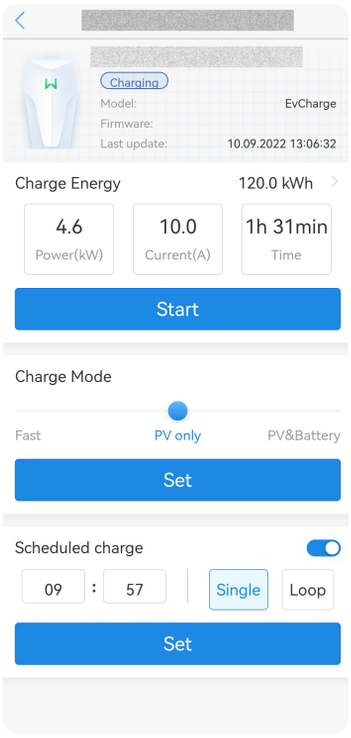
The EV Charger system will charge the EV at maximum power. During day time, pv support the home loads and EV charger at the same time , if pv is not enough, battery will discharge to the home loads and EV charger.

2. PV Priority

The EV charger system will only use the rest pv power to the EV charger. During day time, pv support the home loads first, the rest power will charge the electric vehicle. The battery will not discharge to the EV charger. (Note: If the pv is not enough, grid will charge the electric vehicle)

3. PV&Battery

The EV charger system will user PV power and battery to charge the electric vehicle.



Also, GoodWe smart charging solution also support charge by appointment. Customer can set the time period which electric price is cheap.

With rising energy prices on one side and falling feed-in-tariffs on the other, owners of solar rooftops are well advised to maximize self-consumption and back-up of PV power. By extending the PV and energy storage system to include charging of electric vehicles(EVs) , a high degree of energy independence can be secured. Combine with GoodWe inverters and EV charge for a smart and highly efficient residential ecosystem.

In the near future, GoodWe will add the measure function and the direct communication to GoodWe grid-tied and storage inverter. Let’s wait and see.

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